07-22 第2篇

阅读理解（说明文） Why Does Ice Float on Water?

① In winter, if you leave a bottle of drink in the freezer too long, you know what happens: the bottle breaks. In a cold country, you may see children skating on a frozen lake, while fish swim quietly underneath. At the North Pole, huge sheets of ice float on the sea like white islands. All of these scenes share one strange little fact: **ice, though it is "solid" water, floats on top of liquid water**. Most other things do not do this — put a piece of solid candle wax into melted wax, and it sinks to the bottom. So why is water so special?

② To understand the answer, we need to look very, very closely at water. A tiny drop of water is made of millions of **water molecules**, each one a small "V" shape: one oxygen atom at the point and two hydrogen atoms at the ends. In liquid water, these V-shaped molecules move freely, sliding past each other like children in a busy playground. They can pack quite closely together.

③ When the temperature drops near 0°C, something interesting happens. The water molecules start to slow down, and a weak but steady force between them — called a **hydrogen bond** — has time to pull them into a pattern. Each molecule holds hands with four neighbours, forming a **six-sided ring**, like the shape of a snowflake. Many of these rings join together to make the solid we call ice.

④ Here is the key point: these rings have **gaps** in the middle. The neat, open pattern takes up **more space** than the same number of molecules did when they were moving freely in liquid water. In other words, when water freezes, it does not shrink — it **expands**. This is the opposite of almost all other liquids.

⑤ Because the same amount of water takes up more space as ice, ice is **lighter per cubic centimetre** than liquid water. Anything lighter than water floats, and so does ice. For exactly the same reason, a plastic bottle in the freezer breaks: the water inside pushes outwards on the walls as it freezes and cannot find enough room.

⑥ This odd behaviour of water is not just a fun science fact. It is one of the reasons life on Earth is possible. Imagine for a moment that ice were heavier than water, like most solids. Ice would form at the top of a lake in winter, sink, form again, sink again, and slowly the whole lake would freeze from the bottom up. Fish and plants would not survive. But because ice floats, it covers the lake like a warm blanket. Below this blanket, the water stays a few degrees above freezing. Fish swim, plants live, and spring can come back.

⑦ Scientists have also found practical uses for this property. For example, when we want to keep fresh food cool in a box, putting ice cubes on **top** of the food is more efficient than at the bottom, because cold water sinks and ice stays, making the whole box cool from above downwards. Engineers who build pipes in cold places add extra space or warm layers so that freezing water cannot break them — the same lesson that the freezer bottle teaches us.

⑧ So next time you drop an ice cube into a glass, take a second look. That small, shining block of water is doing something wonderfully unusual — and without its strange habit of floating, the world we know would be a very different, and far colder, place.

根据文章内容，选择最佳答案。

1. What happens when water molecules cool down near 0°C?

A. They become much heavier than before.
B. They slow down, and hydrogen bonds pull them into six-sided rings with gaps inside.
C. They disappear completely.
D. They change into a different element.

2. Why does ice float on water, according to the passage?

A. Because ice is lighter than air.
B. Because frozen water takes up more space than liquid water, so the same amount of water weighs less per cubic centimetre as ice.
C. Because fish push the ice up.
D. Because ice is always warmer than water.

3. Why does a plastic bottle of water break when left in the freezer too long?

A. Because the plastic is too thin for the freezer.
B. Because water expands when it freezes and pushes outwards on the bottle walls.
C. Because the fridge is too hot.
D. Because bacteria eat the plastic.

4. Why is the floating of ice important for life in lakes?

A. Because floating ice looks beautiful in photos.
B. Because ice on top acts like a blanket, keeping water below a few degrees above freezing so fish and plants can survive.
C. Because ice teaches children how to skate.
D. Because fish enjoy cold water only at the bottom of the lake.

5. Which of the following best describes the main idea of the passage?

A. Water freezes faster than any other liquid.
B. Ice floats on water because of the special way water molecules pack when frozen, and this unusual property is very important for life on Earth.
C. Plastic bottles are dangerous in freezers.
D. People in cold countries all know how to skate.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第③段 "hydrogen bond... form a six-sided ring..."
【解析】分子慢下来并以氢键形成六边形环。A/C/D 荒谬。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第④—⑤段 "when water freezes, it does not shrink — it expands... lighter per cubic centimetre..."
【解析】冰体积膨胀、密度更小，故浮起。A/C/D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段 "the water inside pushes outwards on the walls as it freezes..."
【解析】水结冰膨胀撑破瓶子。A/C/D 均不合理。故选 B。

4.
【答案】B
【题型】推理判断题
【定位】第⑥段 "ice covers the lake like a warm blanket..."
【解析】冰像毯子保温，使湖下生命得以存活。A/C/D 均偏。故选 B。

5.
【答案】B
【题型】主旨大意题
【定位】通篇
【解析】全文解释冰为什么浮于水，以及这一特性对地球生命的重要性。A/C/D 均偏。故选 B。

词汇注释（超纲词）

- **molecule** [n.] 分子

- **hydrogen / oxygen** [n.] 氢 / 氧
- **bond** [n.] 键（化学）
- **expand** [v.] 膨胀
- **cubic** [adj.] 立方的